INTERNSHIP REPORT

ENVIROSCAN

AI-Based Pollution Source Identification System using Geospatial Analytics

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INDEX

S.NO	CONTENTS	PAGE NO
1	ABSTRACT	3
2	INTRODUCTION	4
3	PROBLEM STATEMENT	5
4	OBJECTIVES	6
5	SYSTEM ARCHITECTURE	7
6	DATASET DESCRIPTION	10
7	DATA PREPROCESSING	13
8	EXPLORATORY DATA ANALYSIS	15
9	SOURCE LABELING METHODOLOGY	18
10	MODEL DEVELOPMENT	19
11	MODEL EVALUATION	21
12	FEATURE IMPORTANCE ANALYSIS	23
13	GEOSPATIAL VISUALIZATION	24
14	DASHBOARD IMPLEMENTATION	25
15	RESULTS AND DISCUSSION	26
16	LIMITATIONS	29
17	FUTURE WORK	30
18	CONCLUSION	31
19	REFERENCE	32

ABSTRACT

Air pollution is one of the most significant environmental challenges affecting human health and urban sustainability in modern cities. Rapid urbanization, industrial growth, and increasing vehicular emissions have led to a continuous rise in pollution levels. Traditional air quality monitoring systems primarily provide pollutant concentration values but lack the capability to identify pollution sources or offer actionable insights. This limitation reduces their effectiveness in supporting environmental planning and policy-making.

EnviroScan is an AI-based environmental analytics system developed to address these challenges by integrating machine learning with geospatial analysis. The system is designed to analyze air quality data, predict probable pollution sources, and visualize pollution hotspots. It utilizes pollution data collected from the Central Pollution Control Board (CPCB), historical weather data from NASA LARC, and real-time weather data from Open-Meteo. In addition, location-based features such as proximity to roads, industrial zones, dump sites, and farmland are incorporated to simulate environmental conditions and improve predictive performance.

Due to limitations in extracting geospatial features using OpenStreetMap, distance-based location features were synthetically generated. A rule-based labeling approach was applied to classify pollution sources, as real-world labeled datasets were not available. Machine learning models including Random Forest, Decision Tree, and XGBoost were implemented, with XGBoost achieving the highest accuracy in predicting pollution source categories.

The system also integrates geospatial visualization using Folium to generate interactive heatmaps and identify high-risk pollution zones. An interactive dashboard developed using Streamlit enables users to explore pollution data, view predictions, and analyze trends effectively. Overall, EnviroScan demonstrates how data-driven approaches can enhance environmental monitoring by moving beyond traditional systems toward predictive and insight-oriented solutions, thereby supporting better decision-making and sustainable urban development.

Introduction

Air pollution has become a major environmental and public health concern, especially in rapidly urbanizing regions. Increasing industrial activities, vehicular emissions, and population growth have significantly contributed to declining air quality in many cities. Exposure to high levels of pollutants such as PM2.5, PM10, NO_2, and SO_2 can lead to serious health issues, making effective monitoring and analysis essential.

Traditional air quality monitoring systems mainly focus on measuring pollutant concentrations. While these systems provide important data, they do not identify the **sources of pollution** or explain the factors influencing pollution levels. This limitation reduces their effectiveness in supporting targeted mitigation strategies and environmental planning.

With the advancement of artificial intelligence and data analytics, it is possible to enhance environmental monitoring systems by incorporating predictive capabilities. Machine learning models can analyze patterns in pollution data and help identify probable pollution sources. Additionally, geospatial visualization techniques allow better understanding of how pollution is distributed across different locations.

EnviroScan is developed as an AI-based environmental analytics system that integrates pollution data, weather data, and location-based features to provide a comprehensive analysis of air quality. The system uses pollution data collected from CPCB, historical weather data from NASA LARC, and real-time weather data from Open-Meteo. Location-based features such as proximity to roads, industrial zones, and other environmental factors are also considered to improve prediction accuracy.

Due to challenges in extracting geospatial distance features using OpenStreetMap, simulated location features were used to represent real-world conditions. A rule-based labeling approach was adopted to classify pollution sources, as labeled datasets were not available. Machine learning models such as Random Forest, Decision Tree, and XGBoost were implemented, with XGBoost providing the best performance.

The system also includes geospatial visualization using Folium and an interactive dashboard developed with Streamlit, enabling users to explore pollution data, view predictions, and analyze trends effectively.

Overall, EnviroScan demonstrates how integrating machine learning with environmental and spatial data can improve pollution analysis and support informed decision-making for sustainable urban development.

Problem Statement

Air pollution monitoring systems play a crucial role in tracking environmental quality; however, most existing systems are limited to reporting pollutant concentration levels such as PM2.5, PM10, NO_2, and SO_2. While this information is useful, it does not provide insights into the **sources of pollution**, which is essential for implementing effective control measures.

In urban environments, pollution can originate from multiple sources such as vehicular traffic, industrial emissions, waste disposal, and mixed urban activities. Without identifying these sources, it becomes difficult for authorities and stakeholders to take targeted actions to reduce pollution levels. Additionally, traditional systems often lack integration with weather conditions and spatial factors, which significantly influence the dispersion and accumulation of pollutants.

Another key challenge is the absence of labeled datasets that clearly indicate pollution sources, making it difficult to apply supervised machine learning techniques directly. Furthermore, limitations in extracting reliable geospatial features, such as distances to roads and industrial areas, add complexity to the analysis.

Therefore, there is a need for an intelligent system that can integrate pollution, weather, and location-based data to predict probable pollution sources, analyze patterns, and visualize pollution hotspots. Such a system would support better decision-making, improve environmental monitoring, and contribute to sustainable urban management.

Objectives

The primary objective of the EnviroScan project is to develop an intelligent system that can analyze air quality data and identify probable pollution sources using machine learning and geospatial techniques. The system aims to enhance traditional monitoring by providing deeper insights and visual interpretation of pollution patterns.

The key objectives of the project are:

- **To analyze air quality data**
  To study pollution trends using data collected from CPCB and understand variations in pollutant levels across different locations.
- **To predict pollution sources**
  To develop machine learning models that can classify probable pollution sources such as traffic, industrial, and mixed categories based on environmental features.
- **To integrate multiple data sources**
  To combine pollution data, weather data (NASA LARC and Open-Meteo), and location-based features for comprehensive analysis.
- **To perform feature engineering and data preprocessing**
  To clean, transform, and prepare the dataset by handling missing values, merging datasets, and generating meaningful features.
- **To visualize pollution hotspots**
  To use geospatial tools like Folium to generate heatmaps and identify high-risk pollution zones.
- **To develop an interactive dashboard**
  To build a Streamlit-based interface that allows users to explore data, view predictions, and analyze pollution trends.
- **To evaluate machine learning models**
  To compare different models such as Random Forest, Decision Tree, and XGBoost and select the best-performing model.
- **To provide insights for decision-making**
  To support environmental monitoring and planning by presenting clear and actionable insights from the data.

5. System Architecture

The EnviroScan system is designed as a modular architecture that integrates data collection, processing, machine learning, and visualization components to provide a comprehensive environmental analysis platform. The architecture ensures smooth data flow from raw data sources to final user interaction through the dashboard.

The system consists of the following key components:

5.1 Data Collection Layer

This layer is responsible for gathering data from multiple sources:

- Pollution data collected from CPCB (Excel datasets for each station)
- Historical weather data obtained from NASA LARC
- Real-time weather data fetched using Open-Meteo API
- Location-based features generated as simulated datasets due to limitations in OpenStreetMap extraction

This layer ensures availability of diverse environmental data required for analysis.

5.2 Data Processing Layer

In this stage, raw data is cleaned and prepared for analysis:

- Removal of duplicate records
- Handling of missing values using imputation techniques
- Standardization of formats across datasets
- Merging of pollution, weather, and location datasets
- Feature engineering, including distance-based features

This layer transforms raw data into a structured format suitable for modeling.

5.3 Source Labeling Layer

Since real-world labeled data is not available, a rule-based labeling approach is used:

- Pollution sources are assigned based on pollutant thresholds and environmental conditions
- Categories include Traffic, Industrial, and Mixed

This layer generates labeled data required for supervised machine learning.

5.4 Machine Learning Layer

This layer is responsible for model training and prediction:

- Models used: Random Forest, Decision Tree, and XGBoost
- Dataset split into training and testing sets (80:20)
- Model training performed using Kaggle notebook
- Best-performing model (XGBoost) selected and saved

This component enables prediction of pollution sources based on input features.

5.5 Geospatial Visualization Layer

This layer focuses on mapping and spatial analysis:

- Folium is used to generate interactive maps
- Heatmaps are created to visualize pollution intensity
- Monitoring station coordinates are used for plotting

This helps in identifying pollution hotspots and spatial trends.

5.6 Dashboard Layer

The final layer provides user interaction through a web interface:

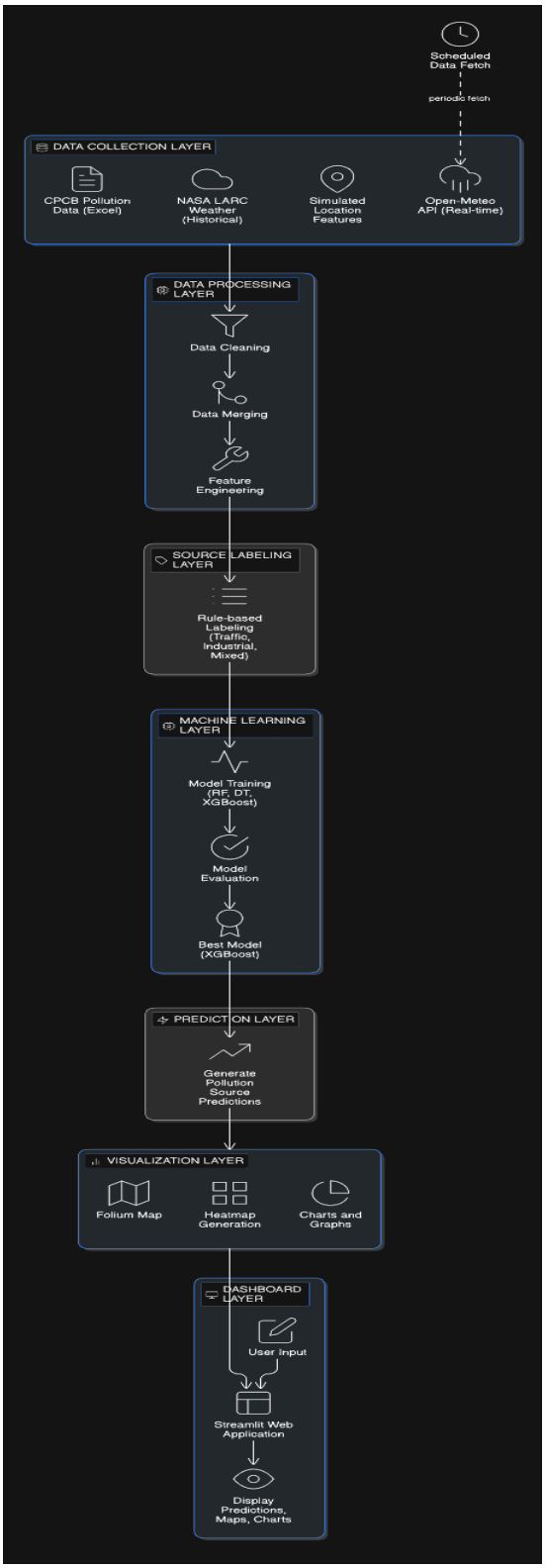
- Built using Streamlit
- Allows user input and data selection
- Displays predictions, charts, and maps
- Provides an intuitive and interactive experience

5.7 Workflow Summary

The overall workflow of the system is as follows:

1. Data is collected from multiple sources
2. Data is cleaned, processed, and merged
3. Pollution sources are labeled using rule-based logic
4. Machine learning models are trained and evaluated
5. Predictions are generated using the trained model
6. Results are visualized using maps and charts
7. Outputs are displayed through the Streamlit dashboard

5.8 System Architecture Diagram



6.Dataset Description

The EnviroScan system utilizes a combination of pollution, weather, and location-based datasets to perform comprehensive environmental analysis. These datasets are collected from multiple sources and integrated to capture both temporal and spatial aspects of air quality. The inclusion of diverse data sources enables the system to better understand pollution patterns and improve prediction accuracy.

6.1 Data Sources

The dataset is constructed using the following sources:

- **Pollution Data (CPCB)**
  Pollution data is collected from the Central Pollution Control Board (CPCB), which provides measurements of key pollutants across multiple monitoring stations.
- **Weather Data**
  Weather conditions play a crucial role in pollution dispersion and are included from two sources:
  - **Historical Data:** NASA LARC dataset
  - **Real-Time Data:** Open-Meteo API
- **Location Data**
  Location-based features were generated as a **simulated dataset** due to challenges in extracting accurate geospatial distance features using OpenStreetMap.

6.2 Monitoring Stations with Coordinates

The dataset includes multiple cities, each with selected monitoring stations. The latitude and longitude of each station are used for geospatial mapping and visualization.

Delhi

- DELHI_CHANDINI_CHOWK (28.6560, 77.2300)
- DELHI_ANANT_VIHAR (28.6469, 77.3153)
- DELHI_ITO (28.6280, 77.2410)

Bhopal

- BHOPAL (23.2336, 77.4009)
- BHOPAL_PARYAVARAN_PARK (23.2500, 77.4200)
- BHOPAL_IDGAH_HILLS (23.2550, 77.4100)

Mumbai

- MUMBAI (19.0470, 72.8746)
- MUMBAI_BYCULLA (18.9767, 72.8331)
- MUMBAI_CHAKALA_ANDHERI (19.1100, 72.8620)

Kolkata

- KOLKATA (22.6270, 88.3800)
- KOLKATA_BALLYGUNGE (22.5250, 88.3630)
- KOLKATA_JADAVPUR (22.4993, 88.3710)

Bengaluru

- BENGELURU (13.0280, 77.5180)
- BENGALURU_HEBBAL (13.0358, 77.5970)
- BENGALURU_RAILWAY_STATION (12.9784, 77.5686)

Chennai

- CHENNAI (13.1660, 80.2580)
- CHENNAI_ALANDUR_BUS_DEPOT (13.0046, 80.2017)
- CHENNAI_VELACHERY (12.9750, 80.2212)

Lucknow

- LUCKNOW (26.8467, 80.9462)
- LUCKNOW_TALKATORA (26.8350, 80.9000)
- LUCKNOW_LALBAGH (26.8485, 80.9416)

Hyderabad

- HYDERABAD (17.5416, 78.4840)
- HYDERABAD_PATENCHERU (17.5330, 78.2600)
- HYDERABAD_SOMAJIGUDA (17.4240, 78.4595)

Ahmedabad

- AHEMEDABAD (23.0300, 72.5400)
- AHMEDABAD_SOMAJIGUDA (23.0370, 72.5666)
- AHMEDABAD_SVPT (23.0350, 72.5800)

6.3 Features

The final dataset includes a combination of pollutant, weather, and location-based features:

- **Pollutant Features**
  PM2.5, PM10, NO2, SO2, CO, Ozone
- **Weather Features**
  Temperature, Humidity, Wind Speed, Wind Direction
- **Location-Based Features**
  - distance_to_major_road_km
  - distance_to_industrial_zone_km
  - distance_to_dump_site_km
  - distance_to_farmland_km

6.4 Summary

The integration of multi-source data enables the EnviroScan system to capture both environmental conditions and spatial context, providing a strong foundation for pollution analysis and prediction.

7. Data Preprocessing

Data preprocessing is a critical step in the EnviroScan system, as it ensures that raw data collected from multiple sources is transformed into a clean, consistent, and structured format suitable for analysis and model training. Since the dataset was obtained from different sources such as CPCB, NASA LARC, Open-Meteo, and simulated location data, preprocessing was necessary to handle inconsistencies and integrate all features effectively.

Each dataset—pollution, weather, and location—was processed individually before being merged into a unified dataset. This step helped in maintaining data quality and ensuring accurate feature representation.

7.1 Preprocessing Steps Performed

- **Cleaning Individual Datasets**
  Each dataset was cleaned separately:
  - Pollution data (CPCB) was processed to remove inconsistencies and standardize pollutant values
  - Weather data (NASA LARC and Open-Meteo) was cleaned and formatted
  - Location features were synthetically generated to represent environmental proximity
- **Handling Missing Values**
  Missing data entries were handled using appropriate techniques such as imputation or removal to ensure dataset completeness.
- **Removing Duplicates**
  Duplicate records were identified and removed to maintain data integrity and avoid bias in model training.
- **Standardization and Formatting**
  Column names, data formats, and units were standardized across all datasets to enable seamless merging.
- **Merging Multiple Datasets**
  The cleaned datasets were combined into a single dataset by aligning common attributes such as:

  - Date
  - City
  - Station
- This resulted in a unified dataset containing pollution, weather, and location features.

- **Synthetic Location Feature Generation**
  Due to challenges in extracting real geospatial distances, location-based features were generated synthetically, including:
  - distance_to_major_road_km
  - distance_to_industrial_zone_km
  - distance_to_dump_site_km
  - distance_to_farmland_km
- **Feature Engineering**
  Additional features were prepared and organized to improve model performance and ensure meaningful input representation.

7.2 Final Dataset Structure

The final merged dataset consists of the following features:

- **Temporal and Location Features**
  - date
  - city
  - station
  - latitude
  - longitude
- **Weather Features**
  - temperature
  - humidity
  - wind_speed
  - wind_direction
- **Pollution Features**
  - pm25
  - pm10
  - no2
  - so2
  - co
  - ozone
- **Location-Based Features**
  - distance_to_major_road_km
  - distance_to_industrial_zone_km
  - distance_to_dump_site_km
  - distance_to_farmland_km

8. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the underlying patterns, relationships, and distributions within the dataset before model development. This step is essential for identifying key trends, detecting anomalies, and gaining insights into how different features influence pollution levels.

The analysis involved examining pollutant distributions, studying relationships between variables, and visualizing data through graphs and plots. These insights helped in feature selection and improved the overall effectiveness of the machine learning models.

8.1 Methods Used

- **Graph-Based Analysis**
  Visualizations such as line plots and bar charts were used to observe pollutant variations across different stations and time periods.
- **Correlation Analysis**
  A correlation matrix was used to identify relationships between pollutants, weather features, and location-based attributes.
- **Trend Analysis**
  Time-series plots were analyzed to study how pollutant levels change over time and across different locations.
-

8.2 Key Observations

- **High PM2.5 in Traffic-Dominated Areas**
  Areas closer to major roads showed elevated PM2.5 levels, indicating the impact of vehicular emissions.
- **Industrial Influence on NO_2 and SO_2**
  Higher concentrations of NO_2 and SO_2 were observed near industrial zones, confirming their role as major pollution sources.
- **Impact of Weather on Pollution Dispersion**
- Weather conditions such as wind speed and humidity influence how pollutants spread, although their direct impact on source classification is limited.

8.3 Visual Analysis

The following visualizations support the observations derived from EDA:

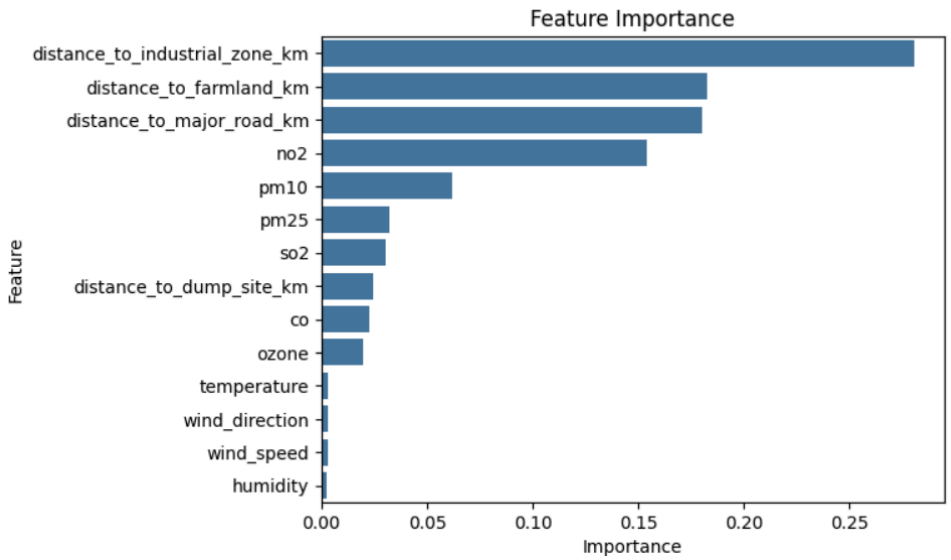


Figure 8.3.1 : Feature Importance Analysis

- Shows that **distance to industrial zones and roads** are highly influential features
- Confirms the importance of spatial features in pollution prediction

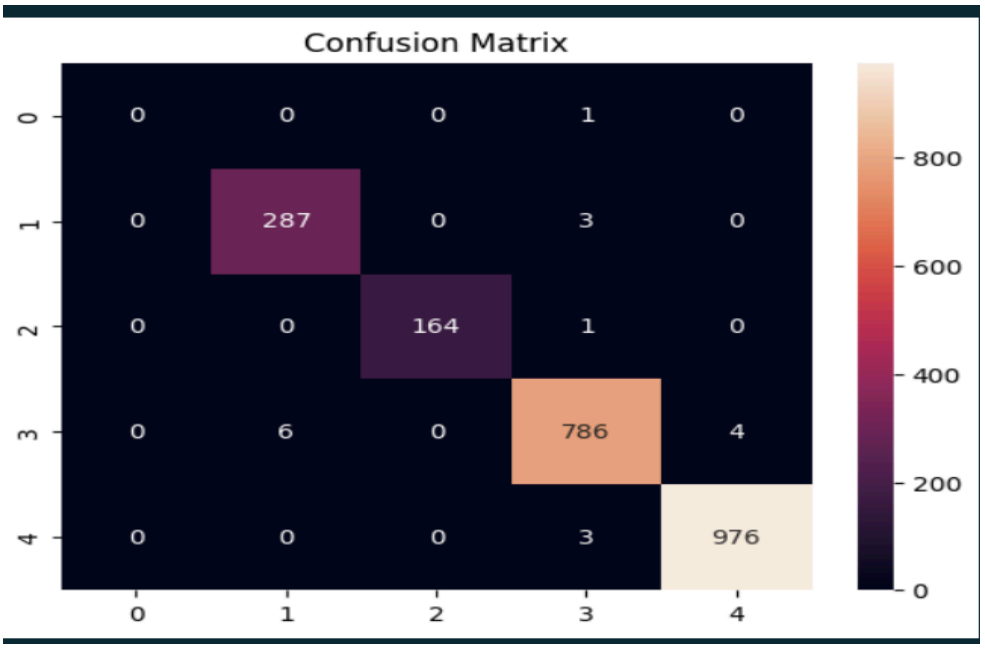


Figure 8.3.2: Confusion Matrix

- Demonstrates the classification performance of the model
- Most predictions lie along the diagonal, indicating correct classification
- Minor misclassifications are observed in less frequent classes

8.4 Summary

EDA provided valuable insights into pollution patterns and feature relationships. It confirmed that **location-based features and pollutant levels are key drivers** in identifying pollution sources. These findings guided the model development process and contributed to achieving high prediction accuracy.

9. Source Labeling Methodology

In the absence of a labeled dataset that explicitly identifies pollution sources, a **rule-based labeling approach** was adopted to assign source categories. This method uses domain

knowledge and predefined thresholds of pollutant values along with location-based features to infer the most probable source of pollution. The approach enables the creation of a labeled dataset required for supervised machine learning.

The labeling process considers both pollutant concentration levels and proximity to environmental factors such as roads and industrial areas. Based on these conditions, each data instance is categorized into a specific pollution source class.

9.1 Labeling Rules

- **Traffic Source**
  High PM2.5 levels combined with close proximity to major roads are classified as traffic-related pollution.
- **Industrial Source**
  Elevated levels of NO_2 and SO_2 are indicative of industrial emissions and are classified accordingly.
- **Mixed Source**
  Data points that do not clearly fall into a single category, or show moderate levels across multiple pollutants, are classified as mixed sources.

9.2 Limitations of Labeling Approach

- **Simulated Labels**
  The labels are generated based on predefined rules rather than real-world verified data, which may introduce inaccuracies.
- **Lack of Ground Truth Validation**
  Since there is no actual dataset specifying pollution sources, the labeling cannot be validated against real-world conditions.
- **Potential Bias in Classification**
  The rule-based approach may oversimplify complex environmental interactions, leading to variations in predicted results.

9.3 Summary

The rule-based labeling methodology provides a practical solution for generating training data in the absence of ground truth. While it enables model development, it also introduces limitations that can be addressed in future work through the use of real labeled datasets and advanced learning techniques.

10. Model Development

The model development phase focuses on building machine learning models to predict pollution sources using the processed and labeled dataset. Multiple classification algorithms were implemented to compare their performance and identify the most suitable model for the task.

The dataset was divided into training and testing sets using an **80:20 split**, ensuring that the models were trained on a majority of the data while maintaining a separate portion for evaluation. Model training was carried out using a **Kaggle notebook environment**, which provided efficient computational resources for experimentation and tuning.

10.1 Models Implemented

- **Random Forest**
  An ensemble learning method that builds multiple decision trees and combines their outputs to improve prediction accuracy and reduce overfitting.
- **XGBoost**
  An advanced gradient boosting algorithm known for high performance and efficiency in handling structured data.

10.2 Development Process

- **Train-Test Split (80:20)**
  The dataset was divided into training and testing subsets to evaluate model performance effectively.
- **Model Training on Kaggle**
  All models were trained using Kaggle notebooks, enabling faster computation and experimentation.
- **Model Comparison and Selection**
  The performance of each model was evaluated based on accuracy and classification metrics, and the best-performing model was selected for final implementation.

Train Test Split

```python
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Training Size:", X_train.shape)
print("Testing Size:", X_test.shape)
```

[9] Python

```
Training Size: (8920, 14)
Testing Size: (2231, 14)
```

Fig 10.2.1 : Train Test Split Results

10.3 Summary

The implementation of multiple models allowed for effective comparison and selection of the most suitable algorithm. Among the models tested, XGBoost demonstrated superior performance and was chosen as the final model for predicting pollution sources.

11. Model Evaluation

The performance of the machine learning models was evaluated using standard classification metrics to assess their effectiveness in predicting pollution sources. Multiple models, including Random Forest and XGBoost, were trained and tested on the processed dataset. The evaluation focused on accuracy and overall classification performance.

11.1 Model Results

- **Random Forest Accuracy:** 0.9914
- **XGBoost Accuracy:** 0.9919

Among the models tested, **XGBoost achieved the highest accuracy** and was selected as the final model for prediction.

11.2 Interpretation of Results

- **High Overall Accuracy**
  Both models demonstrated very high accuracy, indicating that the system is effective in identifying pollution sources based on the given features.
- **Strong Classification Performance**
  The models were able to correctly classify major pollution categories such as Traffic and Industrial with high precision and recall.
- **Slight Imbalance in Rare Classes**
  Some minor classes showed lower representation in the dataset, which may lead to occasional misclassification. This indicates a need for better class balance in future improvements.
- **Model Selection Justification**
  XGBoost was selected due to its slightly higher accuracy and better handling of complex relationships in the data.

Compare Model Accuracy

```python
rf_pred = best_rf.predict(X_test)
xgb_pred = best_xgb.predict(X_test)

rf_acc = accuracy_score(y_test, rf_pred)
xgb_acc = accuracy_score(y_test, xgb_pred)

print("Random Forest Accuracy:", rf_acc)
print("XGBoost Accuracy:", xgb_acc)
```

[16] Python

```
Random Forest Accuracy: 0.9914836396234872
XGBoost Accuracy: 0.9919318691169879
```

Fig 11.3.1 . Comparing Model Accuracy

11.4 Summary

The evaluation results confirm that the selected model performs reliably and is suitable for predicting pollution sources. The high accuracy achieved demonstrates the effectiveness of combining environmental, weather, and location-based features in the modeling process.

12. Feature Importance Analysis

Feature importance analysis was conducted to understand the contribution of different input variables in predicting pollution sources. This analysis helps identify which factors have the most influence on model decisions and provides insights into the relationship between environmental conditions and pollution patterns.

The results indicate that location-based features play a significant role in determining pollution sources, while weather-related features have comparatively lower influence. This is expected, as proximity to pollution-generating activities such as industries and roads directly impacts air quality.

12.1 Key Observations

- **Distance to Industrial Zones (Most Influential Feature)**
  This feature has the highest impact on model predictions, indicating that industrial activities are a major contributor to pollution levels. Areas closer to industrial zones show higher pollution concentrations.
- **Proximity to Major Roads (Significant Impact)**
  Distance to major roads is another important factor, reflecting the influence of vehicular emissions. Locations near high-traffic areas tend to exhibit increased levels of pollutants such as PM2.5 and NO_2.
- **Moderate Contribution of Other Location Features**
  Features such as distance to dump sites and farmland also contribute to predictions but with comparatively lower importance.
- **Lower Impact of Weather Features**
  Weather parameters such as temperature, humidity, and wind speed have a relatively smaller influence on pollution source classification. These features primarily affect pollutant dispersion rather than directly contributing to pollution generation.

12.2 Summary

The feature importance analysis highlights that **spatial proximity to pollution sources** is the most critical factor in predicting pollution categories. This validates the approach of incorporating location-based features in the model and demonstrates the effectiveness of combining environmental and spatial data for pollution analysis.

13. Geospatial Visualization

Geospatial visualization is a key component of the EnviroScan system, enabling the representation of pollution data in a spatial context. It helps in understanding how pollution is distributed across different locations and allows users to easily identify high-risk areas. By integrating location coordinates of monitoring stations with pollution data, the system provides an intuitive visual interpretation of environmental conditions.

The visualization is implemented using **Folium**, a powerful Python library for creating interactive maps. The generated maps allow users to explore pollution patterns dynamically and gain insights into geographical variations in air quality.

13.1 Key Features of Geospatial Visualization

- **Interactive Map Generation**
  The system creates interactive maps using Folium, allowing users to zoom, pan, and explore different regions.
- **Heatmap Visualization**
  Heatmaps are generated to represent pollution intensity, where higher pollution levels are highlighted using stronger color gradients.
- **Pollution Hotspot Identification**
  Areas with high pollution concentration are clearly visible on the map, enabling easy identification of critical zones.
- **Use of Station Coordinates**
  Latitude and longitude values of monitoring stations are used to accurately plot pollution data on the map.
- **Integration with Dashboard**
  The generated maps are integrated into the Streamlit dashboard, allowing users to view spatial data alongside predictions and charts.

Overall, the geospatial visualization component enhances the analytical capability of the system by transforming numerical data into meaningful visual insights. It plays a crucial role in identifying pollution hotspots and supports better understanding of spatial pollution patterns.

14. Dashboard Implementation

The EnviroScan system includes an interactive dashboard developed using **Streamlit**, which serves as the primary interface for user interaction. The dashboard is designed to present complex environmental data in a simple, intuitive, and user-friendly manner. It integrates machine learning predictions, geospatial visualization, and analytical graphs into a single platform, allowing users to explore pollution data effectively.

The dashboard allows users to select monitoring stations and configure parameters such as date and time for analysis. Based on these inputs, the system processes the data and displays the predicted pollution source along with key pollutant metrics. This enables users to understand both the current air quality and the underlying factors contributing to pollution.

14.1 Key Features of the Dashboard

- **User Input Interface**
  Users can select monitoring stations and define analysis parameters, enabling customized data exploration.
- **Prediction Display**
  The system displays the predicted pollution source along with a confidence score, helping users understand the reliability of the model output.
- **Pollutant Metrics Visualization**
  Key pollutant values such as PM2.5, NO_2, SO_2, and CO are presented clearly for quick interpretation.
- **Geospatial Map Visualization**
  An interactive heatmap is generated using Folium to highlight pollution intensity and identify high-risk areas.
- **Trend Analysis Graphs**
  Time-based graphs are provided to analyze variations in pollutant levels over a period.
- **Source Distribution Charts**
  Visual representations such as pie charts show the contribution of different pollution sources.
- **Real-Time Alerts**
  The system provides alerts indicating whether pollution levels are within safe limits or require precautionary measures.

The integration of these features ensures that the dashboard is not only informative but also interactive and easy to use. The use of geospatial mapping combined with analytical charts improves the interpretability of the data, making it easier for users to identify patterns and trends.

Overall, the Streamlit dashboard acts as a unified platform that connects data processing, machine learning predictions, and visualization. Its simple design and accessibility make it suitable for both technical users and general users, thereby enhancing the practical usability of the EnviroScan system.

15. Results and Discussion

The EnviroScan system was successfully developed and tested to analyze air quality data, predict pollution sources, and visualize environmental patterns. The results demonstrate the effectiveness of integrating machine learning with geospatial visualization.

15.1 Pollution Source Prediction

The system is capable of predicting pollution sources based on input features such as pollutant levels, weather data, and location-based attributes. The selected XGBoost model achieved high accuracy and was able to classify sources such as Traffic, Industrial, Mixed, and Natural effectively.

15.2 Dashboard Output

The Streamlit dashboard provides an interactive interface where users can:

- Select monitoring stations
- View predicted pollution source
- Analyze pollutant levels
- Observe real-time alerts

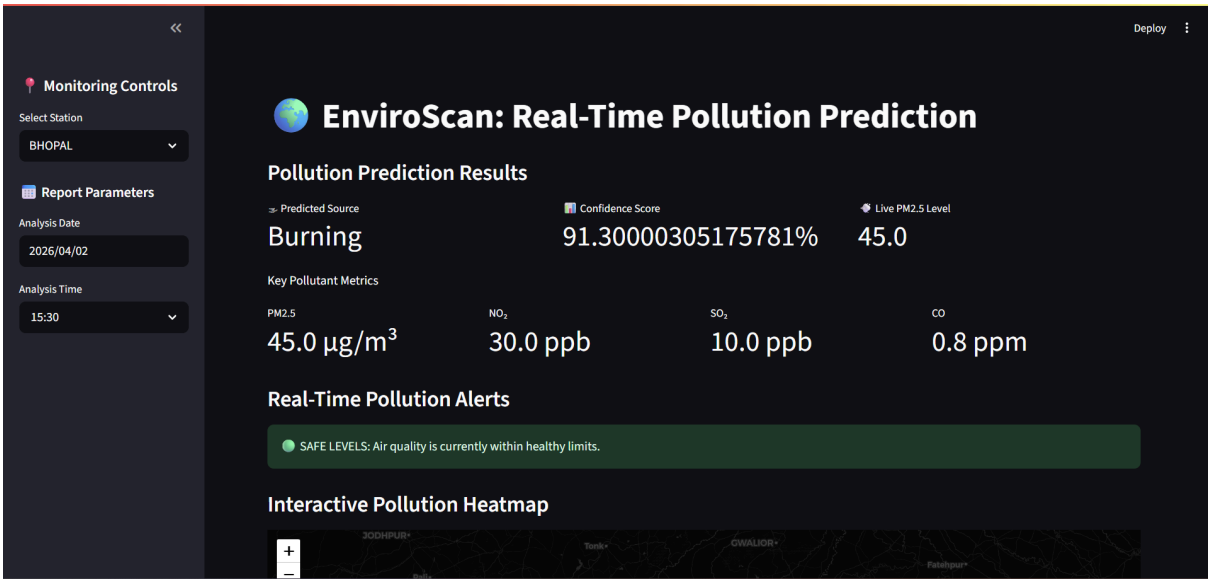


Figure 15.2.1 : Dashboard displaying pollution prediction results and key metrics

15.3 Geospatial Visualization

The system generates interactive heatmaps using Folium to visualize pollution hotspots. These maps help identify high-risk regions based on pollution intensity.

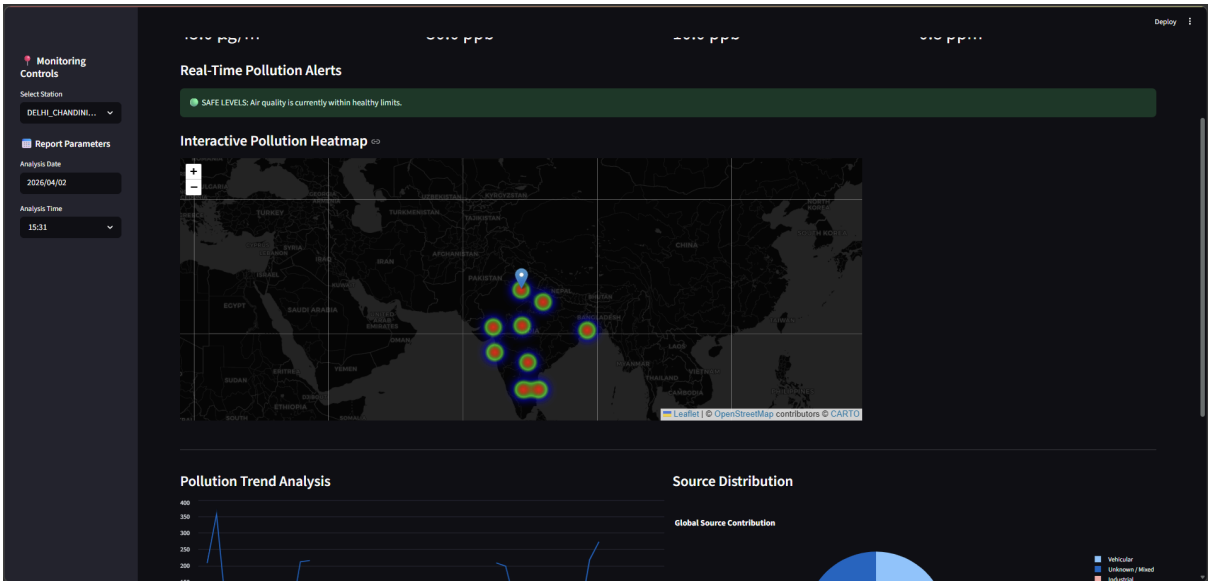


Figure 15.3.1 : Interactive pollution heatmap visualization

15.4 Trend Analysis and Source Distribution

The dashboard includes graphical analysis such as:

- Pollution trend graphs
- Source distribution charts

These visualizations help in understanding how pollutant levels change over time and how different pollution sources contribute.

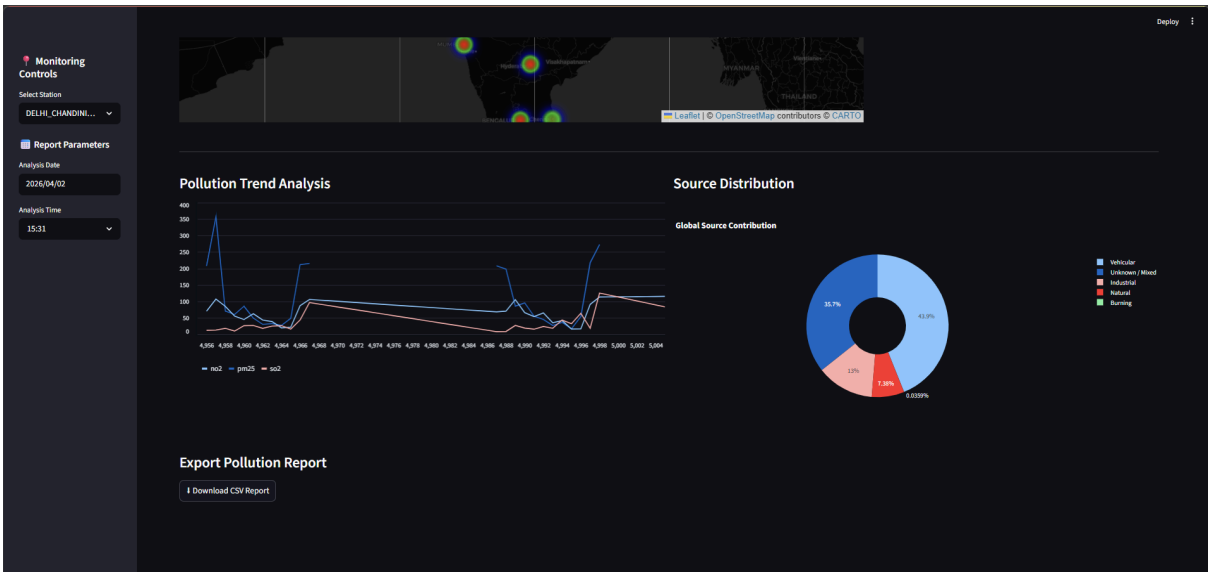


Figure 15.4.1 : Pollution trend analysis and source distribution

15.5 Key Outcomes

The system successfully:

- Predicts pollution sources with high accuracy
- Visualizes pollution hotspots using geospatial maps
- Integrates pollution, weather, and location datasets
- Provides an interactive dashboard for analysis
- Enables better understanding of environmental patterns

16. Limitations

Although EnviroScan provides a functional and insightful approach to pollution analysis, the system has certain limitations due to data constraints and methodological assumptions. These limitations are outlined below:

1. **Simulated Location Data**
   Due to challenges in extracting geospatial distance features using OpenStreetMap, location-based features such as distance to roads, industrial zones, dump sites, and farmland were synthetically generated. As a result, these features may not accurately represent real-world spatial relationships.
2. **Synthetic Distance Features**
   The exact distances from monitoring stations to environmental factors are difficult to obtain using available tools and APIs. Therefore, distance values were artificially created, which may reduce the precision of spatial analysis and model predictions.
3. **Rule-Based Labeling Bias**
   Pollution source labels were assigned using predefined rules based on pollutant thresholds and assumptions. This introduces bias, as real-world pollution patterns may vary across locations and conditions. Consequently, the predicted results may show variations or inaccuracies.
4. **Lack of Ground Truth Data**
   There is no verified dataset that clearly identifies actual pollution sources. The absence of ground truth limits the ability to validate model predictions and affects the reliability of classification.
5. **Limited Real-Time Data Integration**
   While real-time weather data is partially integrated using Open-Meteo, real-time pollution data integration is limited. This restricts the system's ability to provide fully dynamic and up-to-date predictions.

17. Future Work

The current implementation of EnviroScan provides a strong foundation for environmental monitoring; however, several enhancements can be made to improve its accuracy, scalability, and usability. The following future improvements are proposed:

1. **Real Geospatial Data Integration**
   Replace the simulated location-based features with real geospatial data using advanced GIS tools and APIs such as OpenStreetMap or other reliable sources. This will improve the accuracy of distance-based features and make the model more realistic.
2. **Satellite Data Integration**
   Incorporate satellite-based environmental data from sources such as NASA or Sentinel to enhance large-scale pollution analysis. This will enable better detection of pollution spread, land-use patterns, and environmental changes.
3. **Advanced Deep Learning Models**
   Extend the machine learning component by implementing deep learning techniques such as neural networks and time-series models. This can improve prediction accuracy and capture complex relationships in environmental data.
4. **Real-Time Alert and Recommendation System**
   Develop an automated alert system that notifies users when pollution levels exceed safe limits. The system can also provide precautionary suggestions based on predicted pollution sources and AQI levels.
5. **Voice-Based Interaction System**
   Implement a voice-enabled feature where users can ask audio-based queries such as *"What is the AQI today?"*,*"What is the pollution source?"*, or *"What precautions should I take?"*. The system can respond with voice outputs, improving accessibility and user interaction.
6. **React-Based Frontend Development**
   Upgrade the frontend from Streamlit to a more scalable React-based web application. This will improve performance, modularity, and integration with backend APIs.
7. **Low-Literacy Friendly UI Design**
   Design an inclusive user interface that supports users with low literacy levels. This includes simple language, large buttons, icon-based navigation, multilingual support, and visually intuitive layouts.
8. **Scalable Deployment and Cloud Integration**
   Deploy the system on cloud platforms to enable real-time access, scalability, and better performance. This will make the system suitable for large-scale usage across multiple regions.
9. **Automated Model Retraining Pipeline**
   Implement a feedback mechanism where new data can be continuously added to retrain and improve the model over time, ensuring better adaptability to changing environmental conditions.

18. Conclusion

EnviroScan presents a comprehensive approach to environmental monitoring by integrating machine learning, geospatial analytics, and multi-source data processing. The project addresses a key limitation of traditional air quality monitoring systems, which primarily focus on pollutant concentration levels without identifying the underlying sources of pollution. By introducing predictive modeling and spatial visualization, EnviroScan enhances the ability to analyze pollution patterns and derive meaningful insights.

The system successfully combines pollution data from CPCB, weather data from NASA LARC and Open-Meteo, and location-based features to create a unified dataset for analysis. Through effective data preprocessing, feature engineering, and rule-based labeling, the project overcomes challenges such as missing data and the absence of labeled datasets. The use of simulated location features further enables the system to represent real-world environmental conditions despite limitations in geospatial data extraction.

Multiple machine learning models were implemented and evaluated, including Random Forest, Decision Tree, and XGBoost. Among these, XGBoost demonstrated the best performance in predicting pollution source categories with high accuracy. This highlights the effectiveness of ensemble learning techniques in handling complex environmental datasets. The integration of geospatial visualization using Folium allows for the identification of pollution hotspots, providing a clear and intuitive representation of spatial patterns.

The development of a Streamlit-based dashboard enhances the usability of the system by enabling users to interact with the data, view predictions, and analyze trends in a user-friendly manner. This makes the system accessible not only for technical users but also for stakeholders such as policymakers and environmental analysts.

Despite its achievements, the project has certain limitations. The reliance on rule-based labeling introduces potential bias, and the use of simulated location features may not fully capture real-world spatial dynamics. Additionally, real-time data integration and automated model retraining are limited in the current implementation. However, these limitations also highlight opportunities for future improvements.

Overall, EnviroScan demonstrates how data-driven approaches can transform traditional environmental monitoring into a more intelligent, predictive, and insight-oriented system. The project provides a strong foundation for further research and development in the field of environmental analytics. With enhancements such as real-time data integration, advanced modeling techniques, and improved geospatial analysis, the system has the potential to support sustainable urban development and contribute to more effective environmental management strategies.

19. References

The following resources and tools were used during the development of the EnviroScan project:

- **Central Pollution Control Board (CPCB)**
  Source of air quality and pollutant data across monitoring stations in India.
- **NASA LARC (Langley Atmospheric Research Center)**
  Provides historical weather and atmospheric data used for environmental analysis.
- **Open-Meteo API**
  Used for fetching real-time weather data such as temperature, humidity, and wind conditions.
- **OpenAQ Platform**
  Provides access to global air quality data and APIs for environmental monitoring.
- **Scikit-learn Documentation**
  https://scikit-learn.org/
  Used for implementing machine learning models and evaluation techniques.
- **XGBoost Documentation**
  https://xgboost.readthedocs.io/
  Used for implementing gradient boosting models for prediction.
- **Pandas Documentation**
  https://pandas.pydata.org/
  Used for data manipulation and preprocessing.
- **NumPy Documentation**
  https://numpy.org/
  Used for numerical computations and data handling.
- **Folium Documentation**
  https://python-visualization.github.io/folium/
  Used for geospatial visualization and map generation.
- **Streamlit Documentation**
  https://streamlit.io/
  Used for developing the interactive dashboard interface.
- **Kaggle Platform**
  https://www.kaggle.com/
  Used for model training and experimentation.

Summary

These resources played a significant role in data collection, preprocessing, model development, and visualization, contributing to the successful implementation of the EnviroScan system.